Ka Hyun Park

Ph.D. Student at Seoul National University

kahyun.park@snu.ac.kr | Google Scholar

Graph Mining
- Link (Sign) Prediction
- Graph Classification

Speech Analysis
- Automatic Speech Recognition
- Domain Adaptation

Profile

Ph.D. student at Data Mining Lab, Seoul National University, advised by Prof. U Kang. My research focuses on **machine learning under label uncertainty** and **representation learning for graphs**, spanning PU learning, link (sign) prediction, graph classification, and domain adaptation across graph, tensor, and speech data, with publications in top-tier data mining and machine learning venues (e.g., KDD, AAAI, ICDM, TKDE).

Education

Sep 2022 – Present **Seoul National University**
Ph.D. Candidate in Computer Science and Engineering
Data Mining Lab, advised by Prof. U Kang | Expected Feb 2028

Mar 2018 – Aug 2022 **Ewha Womans University**
B.S. in Computer Science and Engineering, Minor in Statistics
GPA 4.14 / 4.50 (Major 4.26 / 4.50)

Publications

* *equal contribution.* ***Bold*** *indicates the author.*

2026 **J3** Fast and Accurate Domain Adaptation for Irregular and Regular Tensor Decomposition
Junghun Kim, ***Ka Hyun Park****, Jun-Gi Jang, and U Kang*
IEEE Transactions on Knowledge and Data Engineering (TKDE), 2026.

2025 **C6** PiGLeT: Probabilistic Message Passing for Semi-supervised Link Sign Prediction
Ka Hyun Park, *Junghun Kim, Jinhong Jung, and U Kang*
IEEE International Conference on Data Mining (ICDM), 2025.

J2 Accurate Semi-supervised Automatic Speech Recognition for Ordinary and Characterized Speeches via Multi-hypotheses-based Curriculum Learning
Ka Hyun Park*, *Junghun Kim*, and U Kang*
PLOS ONE, 2025.

C5 Accurate Graph-based Multi-Positive Unlabeled Learning via Disentangled Multi-view Feature Propagation
Junghun Kim, Hoyoung Yoon, ***Ka Hyun Park****, and U Kang*
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2025.

C4 Accurate Link Prediction for Edge-Incomplete Graphs via PU Learning
Junghun Kim, ***Ka Hyun Park****, Hoyoung Yoon, and U Kang*
AAAI Conference on Artificial Intelligence (AAAI), 2025.
⋆ Selected as Oral Presentation.

2024 **C3** Domain-Aware Data Selection for Speech Classification via Meta-Reweighting
Junghun Kim, ***Ka Hyun Park****, Hoyoung Yoon, and U Kang*
Interspeech, 2024.

C2 Fast and Accurate Domain Adaptation for Irregular Tensor Decomposition
*Junghun Kim, **Ka Hyun Park**, Jun-Gi Jang, and U Kang*
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2024.

C1 Accurate Semi-supervised Automatic Speech Recognition via Multi-hypotheses-based Curriculum Learning
Junghun Kim, **Ka Hyun Park***, and U Kang*
Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2024.
⋆ Selected as Oral Presentation.

2023 **J1** Accurate Graph Classification via Two-Staged Contrastive Curriculum Learning
*Sooyeon Shim, Junghun Kim, **Ka Hyun Park**, and U Kang*
PLOS ONE, 2023.

Under Review

R1 Accurate Source-Free Speech Classification via Meta-Learned Target-Centric Model Merging
Ka Hyun Park*, *Junghun Kim*, and U Kang*

R2 Accurate Graph Classification via Contextualized Structural Tokens
Ka Hyun Park, *Junghun Kim, and U Kang*

Patents

Filed = application filed. Reg. = registered/granted. KR = Korea, US = United States, JP = Japan, EP = Europe.

2026 (US, Filed) Disentangled Multi-View Graph Propagation Method and Apparatus for Multi-Positive and Unlabeled Node Learning (D-MVP)

2025 (KR, Filed) Semi-supervised Link Sign Prediction Based on Probabilistic Message Passing (PiGLeT)

(KR, Filed) Disentangled Multi-View Graph Propagation for Multi-Positive and Unlabeled Node Learning (D-MVP)

(JP, Reg.) Link Prediction Using Accurate Edge Prediction Model Based on Positive-Unlabeled Data Learning (PULL)

2024 (KR, Filed) Speech Classification Apparatus and Method via Meta-Reweighting (DoReMe)

(EP, Filed) Link Prediction Based on Positive-Unlabeled Data Learning (PULL)

(US, Filed) Link Prediction Based on Positive-Unlabeled Data Learning (PULL)

(KR, Reg.) Link Prediction Using Accurate Edge Prediction Model Based on Positive-Unlabeled Data Learning (PULL)

(KR, Reg.) Tensor Decomposition Apparatus and Method for Domain Adaptation (Meta-P2)

2023 (KR, Filed) Semi-supervised Automatic Speech Recognition Using Multi-Hypotheses-Based Curriculum Learning (MOCA)

(KR, Filed) Graph Classification via Contrastive Curriculum Learning (TAG)

Talks

Dec 2025	PiGLeT: Probabilistic Message Passing for Semi-supervised Link Sign Prediction KSC Top Conference Session

Teaching Assistant

2026	Data Mining — K-MOOC
2024, 2023	Data Structure (M1522.000900) — Seoul National University
2023	DS^2 Deep Learning — Samsung

Reference

Prof. U Kang, Ph.D. Advisor — Seoul National University: ukang@snu.ac.kr